

$$G = \mathrm{SL}_2(\mathbb{F}_q)$$

$$Y = \{xy^q - yx^q = 1\} \subset G \times \mu_{q+1}$$

$K/\mathbb{Q}_\ell$ fin ext containing all $|G|$ th roots of 1.

$$\leadsto H_c^i(Y) := H_c^i(Y, K) \text{ a } (KG, K\mu_{q+1})\text{-bimod}$$

$$\leadsto \text{For } \Theta \in K_0(K\mu_{q+1})$$

$$R'(\Theta) = [H_c^*(Y) \otimes_{K\mu_{q+1}} \Theta]_G \in K_0(KG)$$

If Θ is a character, i.e. trace of a rep V_Θ then

$$R'(\Theta) = [H_c^1(Y) \otimes_{K\mu_{q+1}} V_\Theta]_G - [H_c^2(Y) \otimes_{K\mu_{q+1}} V_\Theta]_G$$

and by A.2.1 (c)

$$[H_c^2(Y)]_{G \times \mu_{q+1}} = 1_{G \times \mu_{q+1}}$$

Cor If $\Theta: \mu_{q+1} \rightarrow K^\times$ is a nontrivial linear char, then

$$R'(\Theta) = [H_c^1(Y) \otimes_{K\mu_{q+1}} K_\Theta]_G, \quad K_\Theta = K \text{ with } \mu_{q+1} \text{ acting by } \Theta$$

and this is the character of a rep of G .

Notation If Θ is any class function on a fin group Γ , let Θ^* be the class function $\Theta^*(\gamma) = \Theta(\gamma^{-1})$.

Prop For $\Theta \in K_0(K\mu_{q+1})$,

$$R'(\Theta) = R'(\Theta^*) = R'(\Theta)^*$$

Proof Let $\varphi: \mu_{q+1} \rightarrow \mu_{q+1}$ be $\varphi(\gamma) = \gamma^{-1}$, so $\Theta^* = \Theta \circ \varphi$.

We first want to show

$$R'(\Theta) = R'(\Theta^*) = R'(\Theta \circ \varphi)$$

Let $\varphi|_{H_c^i(Y)}$ be the $(KG, K\mu_{q+1})$ -bimod where the μ_{q+1} -action is precomposed with φ . To show $R'(\Theta) = R'(\Theta^*)$, it is enough to show

$$H_c^i(Y) \cong \varphi|_{H_c^i(Y)} \text{ as } KG\text{-mods}$$

But the Frobenius

$$F: H_c^i(Y) \rightarrow H_c^i(Y) \text{ is a } KG\text{-mod iso}$$

and satisfies $F \circ \gamma = \gamma^q \circ F = \gamma^{-1} \circ F$ on Y , for $\gamma \in \mu_{q+1}$
 $\Rightarrow F: H_c^i(Y) \xrightarrow{\varphi} H_c^i(Y)$

Next we want to show $R'(\Theta^*) = R'(\Theta)^*$.

First note,

$$K\mu_{q+1} = \bigoplus_{\alpha: \mu_{q+1} \rightarrow K^\times} K_\alpha \text{ with } e_\alpha K\mu_{q+1} \text{ where } e_\alpha = \frac{1}{q+1} \sum_{\gamma \in \mu_{q+1}} \alpha(\gamma^{-1}) \gamma$$

Then for $\Theta: \mu_{q+1} \rightarrow K^\times$ a lin char,

$$R'(\Theta) = H_c^*(Y) e_\Theta$$

Then for $g \in G$,

$$R'(\Theta)(g) = -\frac{1}{q+1} \sum_{\gamma \in \mu_{q+1}} \mathrm{Tr}_Y^*(g, \gamma) \Theta(\gamma^{-1})$$

$$\text{Thus } R'(\Theta)^*(g) = R'(\Theta)(g^{-1})$$

$$= -\frac{1}{q+1} \sum_{\gamma \in \mu_{q+1}} \mathrm{Tr}_Y^*(g^{-1}, \gamma) \Theta(\gamma^{-1})$$

$$= -\frac{1}{q+1} \sum_{\gamma \in \mu_{q+1}} \mathrm{Tr}_Y^*(g^{-1}, \gamma) \Theta^*(\gamma)$$

$$\text{A.2.6} \Rightarrow \mathrm{Tr}_Y^*(g^{-1}, \gamma) \in \mathbb{Z}$$

$$\Rightarrow \mathrm{Tr}_Y^*(g^{-1}, \gamma) = \mathrm{Tr}_Y^*(g, \gamma^{-1})$$

$$\Rightarrow R'(\Theta)^*(g) = -\frac{1}{q+1} \sum_{\gamma \in \mu_{q+1}} \mathrm{Tr}_Y^*(g, \gamma^{-1}) \Theta^*(\gamma) = R'(\Theta^*)(g) \quad \square$$

Let's consider the special case $\Theta = 1$.

Then

$$R'(1) = [H^1(Y)^{\mu_{q+1}}]_G - [H^2(Y)^{\mu_{q+1}}]_G = [H^1(Y)^{\mu_{q+1}}]_G - 1_G \text{ by A.2.1 (c)}$$

On the other hand since

$$Y/\mu_{q+1} \cong \mathbb{P}^1(\mathbb{F}) \setminus \mathbb{P}^1(\mathbb{F}_q) \quad (\text{Sept 16})$$

we have

$$R'(1) = [H_c^*(Y)^{\mu_{q+1}}]_G = [H_c^*(Y/\mu_{q+1})]_G \quad \text{A.2.4}$$

$$\begin{aligned}
&= [H_c^*(P'(F) \setminus P'(F_q))]_G \\
&= [H_c^*(P'(F))]_G - [H_c^*(P'(F_q))]_G
\end{aligned}$$

Now

$$\begin{aligned}
[H_c^*(P'(F))]_G &= [H^0]_G - [H^1]_G + [H^2]_G \\
&= 1_G - 0 + 1_G = 2 \cdot 1_G \text{ by last lecture}
\end{aligned}$$

and

$$[H_c^*(P'(F_q))]_G = [K[G/B]]_G = 1_G + 5t_G$$

Putting this all together

$$R'(1) = 5t_G - 1_G$$

and

$$[H_c^i(Y) \otimes_{K/\mu_{q+1}} 1_{\mu_{q+1}}]_G = \begin{cases} 1_G & i=2 \\ 5t_G & i=1 \\ 0 & \text{else} \end{cases}$$